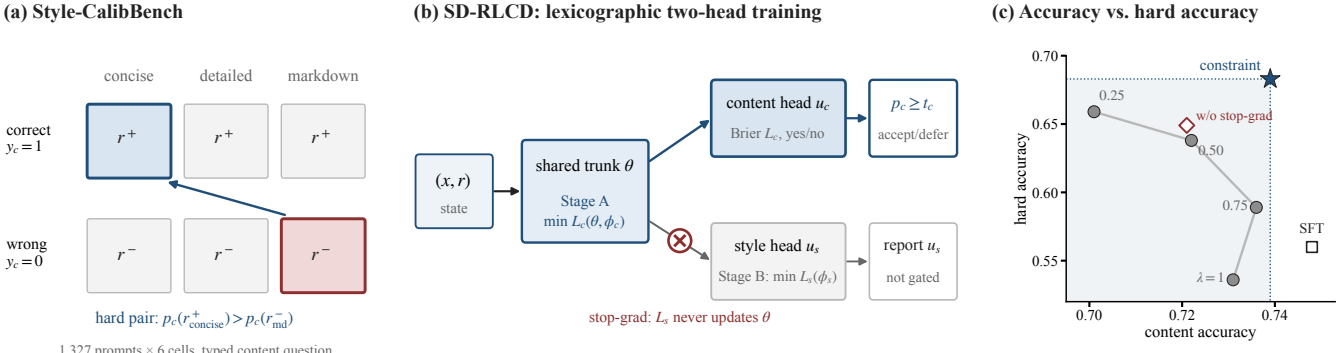

(a) Style-CalibBench
concise
detailed
markdown
correct
$y_c = 1$
wrong
$y_c = 0$
r^+
r^+
r^+
r^-
r^-
r^-
hard pair: $p_c(r^+_{\text{concise}}) > p_c(r^-_{\text{md}})$
1,327 prompts × 6 cells, typed content question
(b) SD-RLCD: lexicographic two-head training
(x, r)
state
shared trunk θ
Stage A
min $L_c(\theta, \phi_c)$
content head u_c
Brier L_c, yes/no
$p_c \geq t_c$
accept/defer
style head u_s
Stage B: min $L_s(\phi_s)$
report u_s
not gated
stop-grad: L_s never updates θ
(c) Accuracy vs. hard accuracy
hard accuracy
content accuracy
0.70
0.65
0.60
0.55
0.70
0.72
0.74
constraint
w/o stop-grad
SFT
0.25
0.50
0.75
$\lambda = 1$